

Random Events

Learning Objectives

- Use random number generation in Scratch programs
- Create unpredictable game elements using randomness
- Understand how randomness adds variety and replayability

Engage (5-7 minutes)

Nature is unpredictable — rain may fall unexpectedly, a cow may wander to a different field, a bird may land anywhere. Random events in Scratch bring this unpredictability to programs, making games more interesting like the natural variety farmers experience daily.

Explore (10-12 minutes)

Students experiment with “pick random” blocks for positions, directions, and values. They see how random numbers create varied outcomes each time the program runs. Like dice rolling during village gatherings, randomness brings chance and surprise to their programs.

Explain (10-12 minutes)

The “pick random” block generates numbers within a specified range. Using random positions makes sprites appear unpredictably. Random directions create varied movement patterns. Random timing adds suspense. Like the unpredictable monsoon rains that bring both joy and challenge, randomness makes programs dynamic and engaging.

Elaborate (8-10 minutes)

Students create a game where coins appear at random positions for the player to collect. Each playthrough is different because positions are randomized, like harvesting different amounts of produce each season — never exactly the same twice.

Evaluate (5-8 minutes)

Can students implement random number generation in their projects? Can they explain why randomness improves games? Do they use random values in at least three different ways?